VERSION 5.0 / FINAL CHANGE REPORT

Reports, security & final polish

Version 5 completes the CSTBS release sequence with administrator PDF/CSV exports, request hardening, demonstration workflows and comprehensive manuals.

RELEASE	GITHUB TAG	SYSTEM
Final development milestone	v5.0	Flask / SQLAlchemy / SQLite / ReportLab

What Version 5 adds

Booking report generation

Admin-only PDF and CSV reports filtered by event and reservation status. Includes owner, event, seat, created date, status and booked price.

Event reports

Event and venue details plus seat totals, reserved seats, blocked seats and remaining capacity. Live SQL aggregation keeps counts current.

Safer exports

Landscape PDFs with repeating headers and page numbers; UTF-8 CSV with spreadsheet formula protection; private, non-cacheable attachments and output limits.

Security upgrades

Flask-Limiter rate limits, 12-character minimum registration passwords, bounded form values, CSRF expiry, size limits, security headers and SQLite foreign keys.

Deployment structure

Explicit init-db command, production configuration checks, HTTPS session settings, shared limiter support and a local-only non-debug development server.

Demonstration and documentation

Optional opt-in sample accounts and bookings, refreshed admin reporting UI, User Manual, Admin Manual, deployment guide and security guide.

Retained from Versions 1–4

Responsive event catalog, Flask-Login accounts and roles, full event CRUD, interactive seat selection, atomic bookings, cancellation/history, dashboard, seat and section administration, all-user booking oversight and Flask-Mail notifications.

Scope: No payments, issued tickets or refunds. SQLite, synchronous email and five-second polling are sized for small installations and demonstrations.

ARCHITECTURE / SECURITY

Report flows and safeguards

Both report types reuse existing User, Event, Seat and Booking tables; no Version 5 database migration is required. Exports execute read-only queries and write directly into downloadable in-memory streams.

Admin export routes

Route	Output and filters
/admin/reports/	Role-protected report center with booking and event forms.
/admin/reports/bookings.csv	CSV booking history; optional event/status filters.
/admin/reports/bookings.pdf	Paged landscape PDF booking history; same filters.
/admin/reports/events.csv	CSV event and seating inventory counts.
/admin/reports/events.pdf	Paged landscape PDF event and seating report.

Defense in depth

- Only logged-in admin users can reach report endpoints; students/staff receive HTTP 403, guests redirect to login.
- Sensitive files are attachments with private, no-store response headers; generated reports are not placed in the public static directory.
- CSV input that begins with =, +, -, @ (after whitespace) is prefixed to prevent spreadsheet formula execution.
- Exports are capped at 2,000 PDF or 10,000 CSV rows; report filters are explicitly validated.
- Flask-WTF CSRF protects writes; Flask-Limiter restricts auth POSTs, availability polling and booking confirmation.
- Production mode requires strong secrets, trusted hosts, HTTPS-only session cookies, shared rate-limit storage and non-console SMTP.

Preserved booking invariants

A partial SQLite unique index prevents two active bookings for the same seat. Triggers prohibit booking blocked seats or blocking reserved seats. Multi-seat confirmations are all-or-nothing, and cancellation retains historical records.

Privacy: Booking exports contain names and email addresses. Access, storage, distribution and deletion of downloaded reports require appropriate campus policies.

DELIVERY / OPERATIONS

Installation and GitHub release

Local setup in seven steps

- 1. Extract CSTBS-v5.zip and open the project directory.
- 2. Use Python 3.10+ and activate a virtual environment.
- 3. Install python -m pip install -r requirements.txt.
- 4. Copy .env.example to .env and replace SECRET_KEY with a random generated secret.
- 5. Run python -m flask --app run.py init-db and python -m flask --app run.py seed-events.
- 6. Start python run.py, visit http://127.0.0.1:5000 and register.
- 7. Promote a trusted user with python -m flask --app run.py set-role email@campus.edu admin, then use /admin/reports/.

Complete documentation delivered

File	Contents
docs/user-manual.md	Accounts, event browsing, booking, cancellation and troubleshooting.
docs/admin-manual.md	Admin dashboard, events, seats, roles, all bookings and reporting.
docs/installation-deployment.md	Environment, backups, HTTPS/Redis/SMTP, Gunicorn and operational checks.
SECURITY.md + CHANGELOG.md	Threat notes, configuration checks, and the entire v1.0–v5.0 history.

Validation status

Offline checks: Python syntax, Jinja template compilation, static assets, SQLite constraint checks, PDF rendering and source ZIP integrity. Runtime checks: run the included Flask integration test suite and deploy-only SMTP/Redis checks locally. Package installation was unavailable in this runner.

Test command: python -m unittest discover -s tests -v

GitHub release

Title: CSTBS v5.0 - Final Release: Reporting, Security & Documentation

Tag: v5.0 | Commit suggestion: feat: finalize CSTBS v5.0 reporting and deployment documentation

Version 5 adds administrator booking/event PDF and CSV exports, stricter validation and rate limiting, deployment-oriented configuration, optional local demo data, and comprehensive user/admin/deployment manuals, while preserving all Version 1–4 features.